LAB 3 – ARCHITECTURAL JUSTIFICATION

Multi-Cloud Asset Cost Optimization Portal

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SUBJECT: SE LAB-3

Architecture Selection: We selected Microservices Architecture for the Multi-Cloud Asset Cost Optimization Portal.

Architectural Choice: The reason for choosing the Microservices Architecture is that the portal comprises different components such as billing ingestion, cost analysis, asset optimization, creation of dashboards and budget alerts that can be independently managed.

Reason 1 – Independent Scalability: Billing ingestion and analysis can be independently scaled to deal with multi-gigabyte cloud billing files and to meet the requirement of updating dashboards within 2 minutes.

Reason 2 – Fault Isolation: If there is an issue in any particular service, such as the alerting service or optimization service, the fault will be isolated and other services can work normally.

Advantage of Security: Different services can have API communication, which is controlled and authenticated, to control access to billing and optimization information between services.

Advantage of Performance: The independent scalability of the billing ingestion and analysis services can increase processing performance in handling cloud billing files.

Conclusion: The Microservices Architecture is the appropriate choice for the portal due to its ability to support modularity, independent scalability, fault isolation, security-controlled communication, and performance.

